

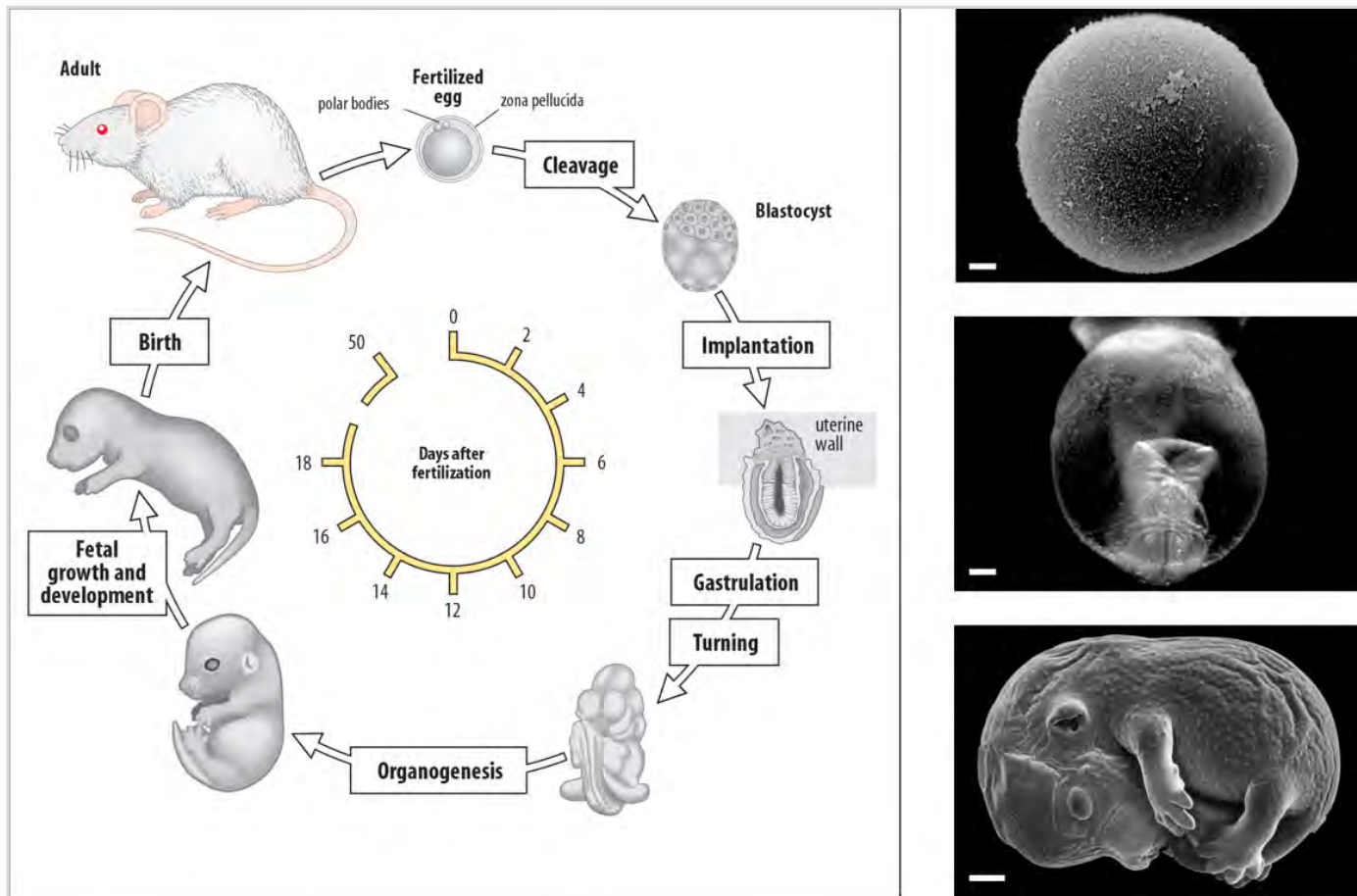
Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

What we covered

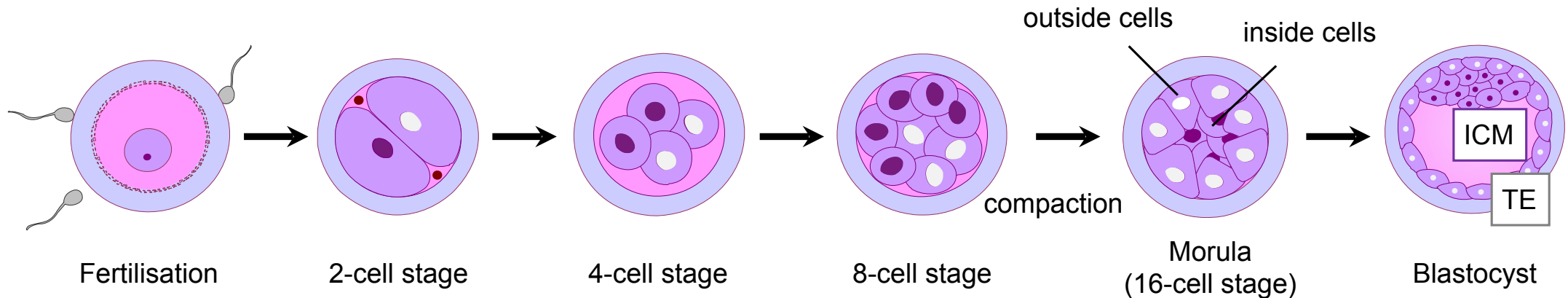
- Quantitative imaging of transcription factor dynamics in mouse (mus musculus) embryos
-

The life cycle of the mouse



How does a single cell develop into a complex multicellular organism?

Early mammalian embryo development



Models of asymmetry generation

‘Stochastic’ model

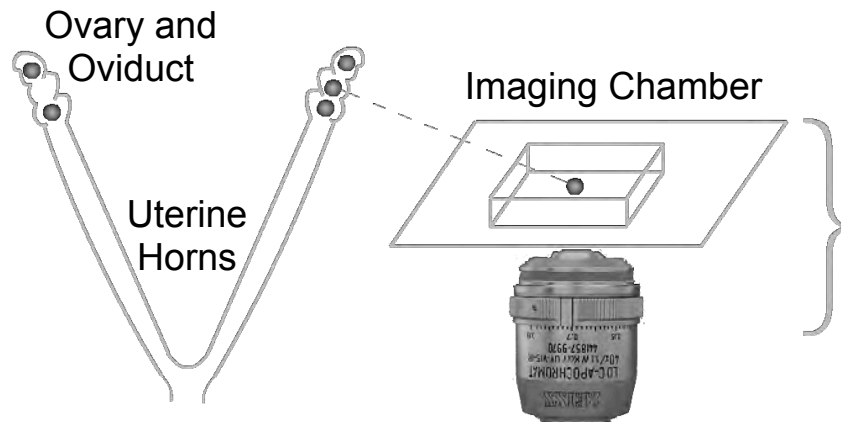
First asymmetry at the 16-cell stage.

‘Biased heterogeneity’ model

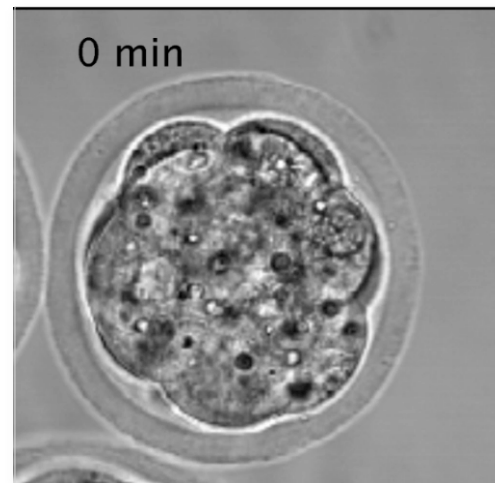
Bias does exist before compaction.

Only static readouts

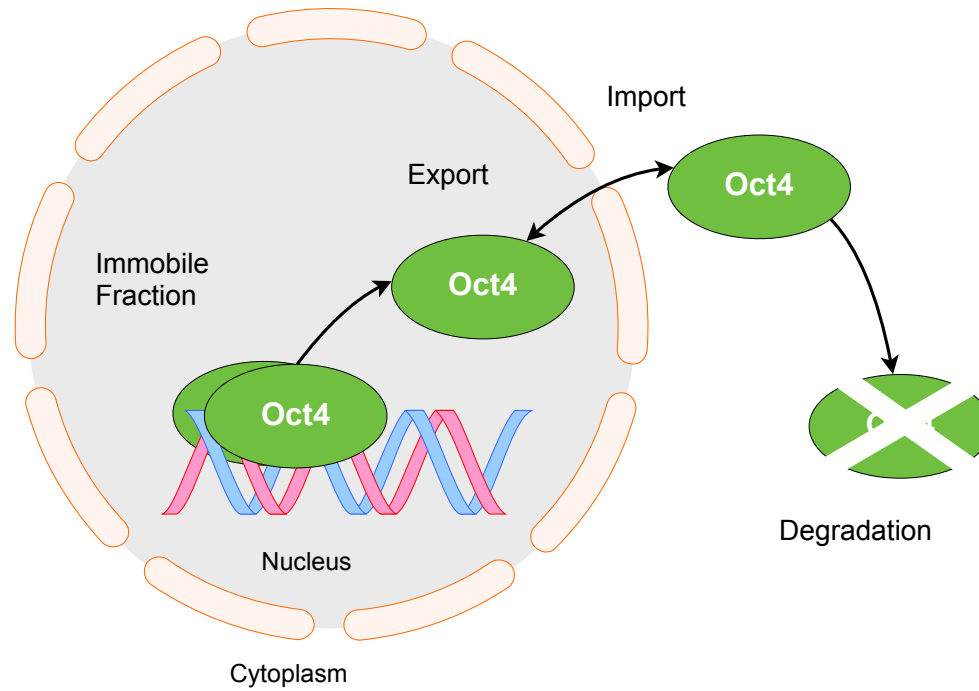
Imaging mouse embryos



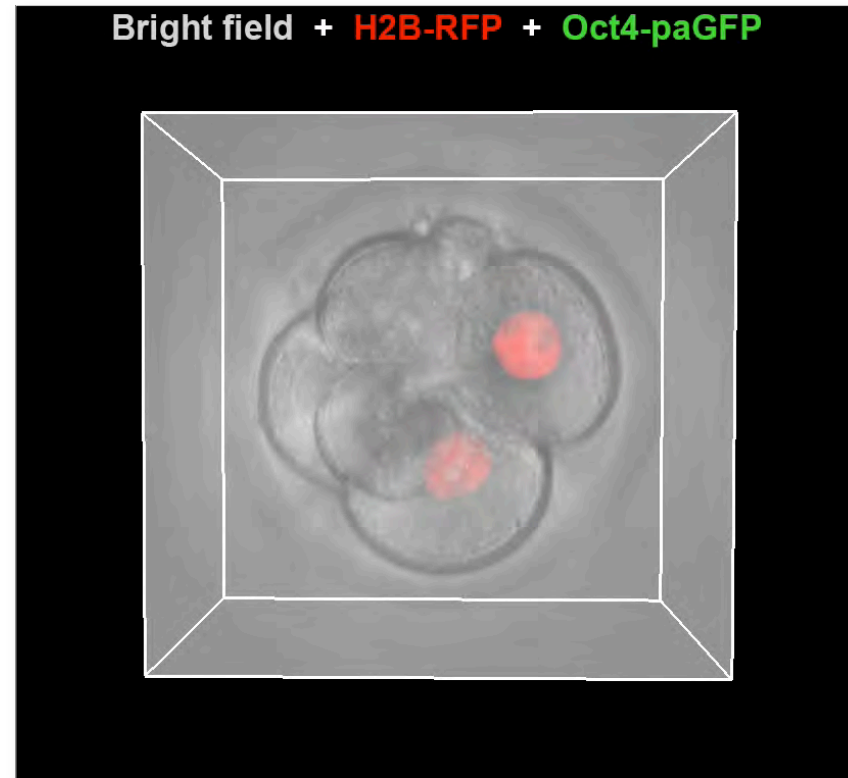
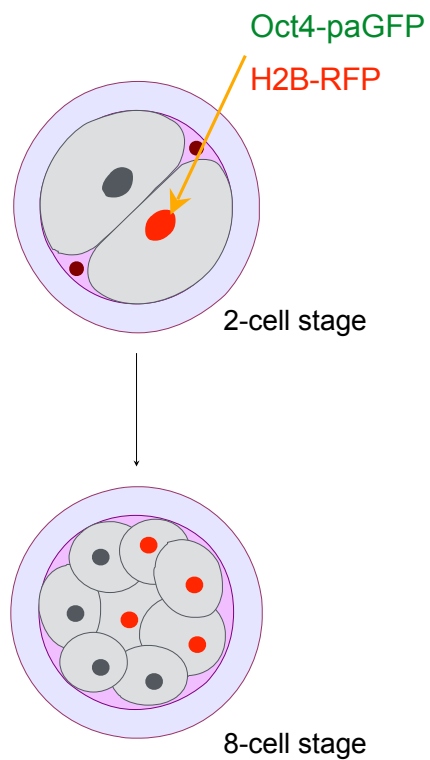
Laser Scanning Microscope



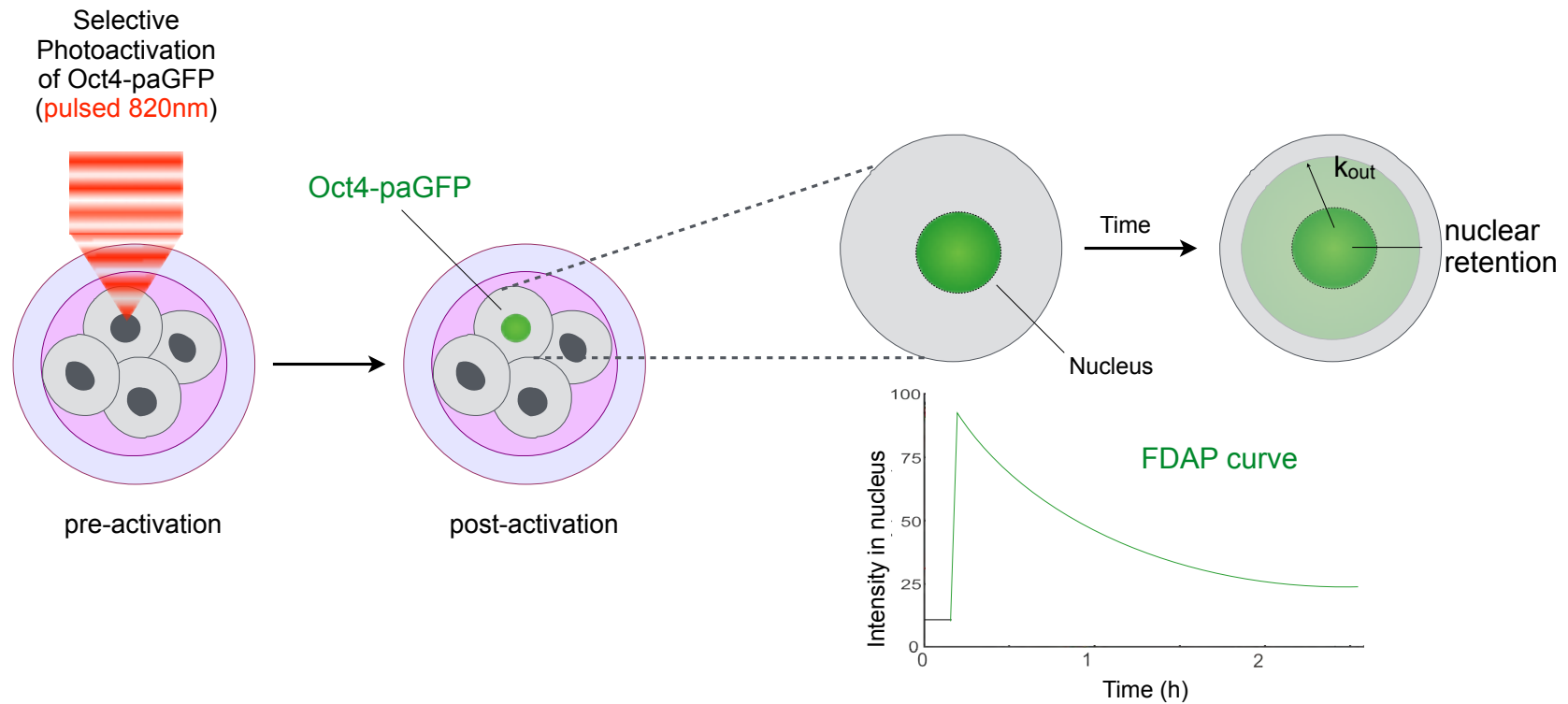
Transcription factor dynamics: Export-Import-Degradation-Immobile fraction



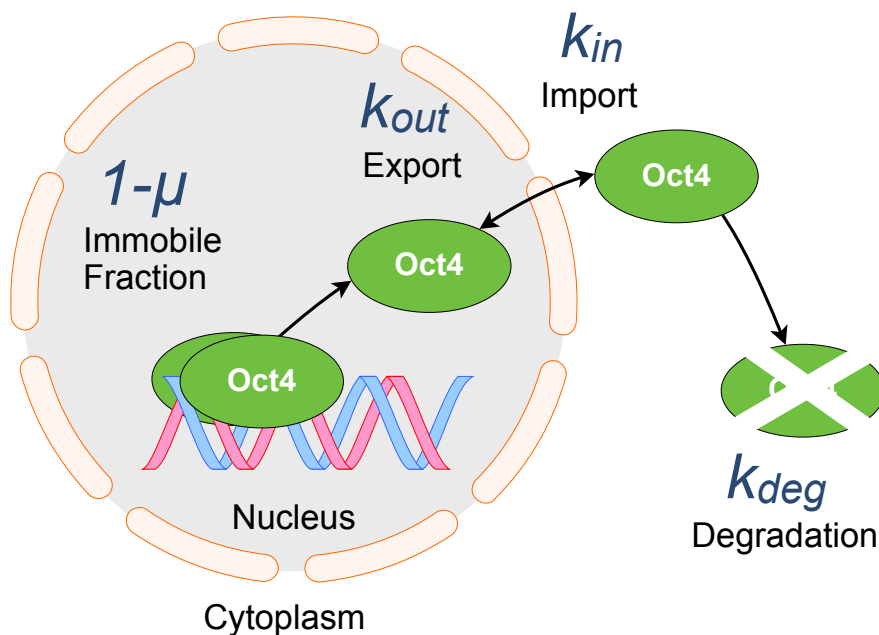
Oct4-paGFP photoactivation in single cells of the mouse embryo



Fluorescence Decay After Photoactivation (FDAP) of photoactivatable Oct4



Transcription factor dynamics: Export-Import-Degradation-Immobile fraction



Description of Oct4-paGFP dynamics

$$\frac{d}{dt} Oct4_n = k_{in} Oct4_c - (k_{out} + k_{deg}) Oct4_n$$

Export rate k_{out} & immobile fraction $1-\mu$

$$f(t) = \mu f_0 e^{-(k_{out} + k_{deg})t} + (1 - \mu) f_0$$

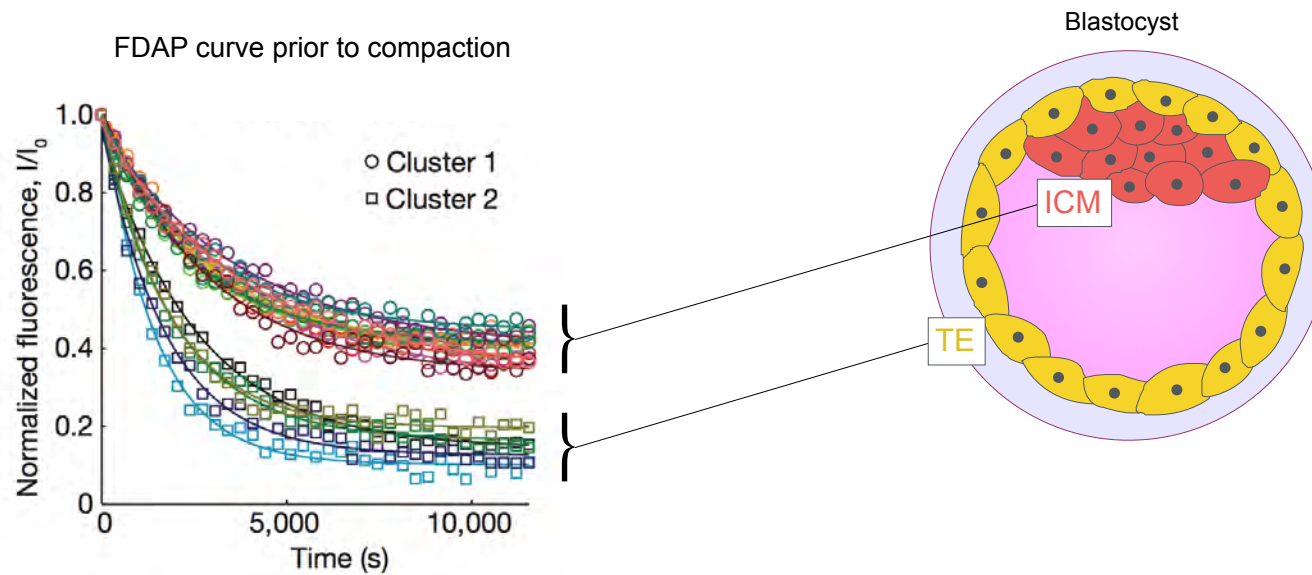
Degradation k_{deg}

$$f(t) = f_0 e^{-k_{deg} t}$$

Import rate k_{in}

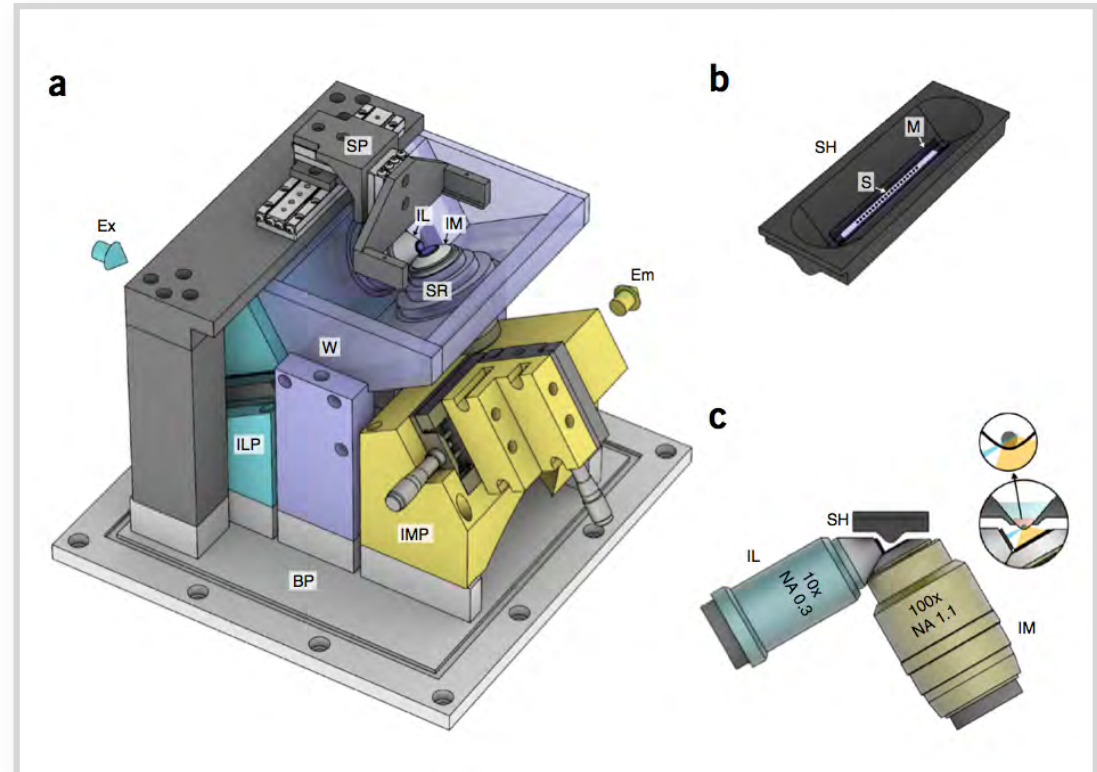
$$k_{in} = (k_{out} + k_{deg}) \mu \frac{Oct4_n^{total}}{Oct4_c}$$

Oct4 kinetics predict ICM/TE lineage patterning in the early mammalian embryo

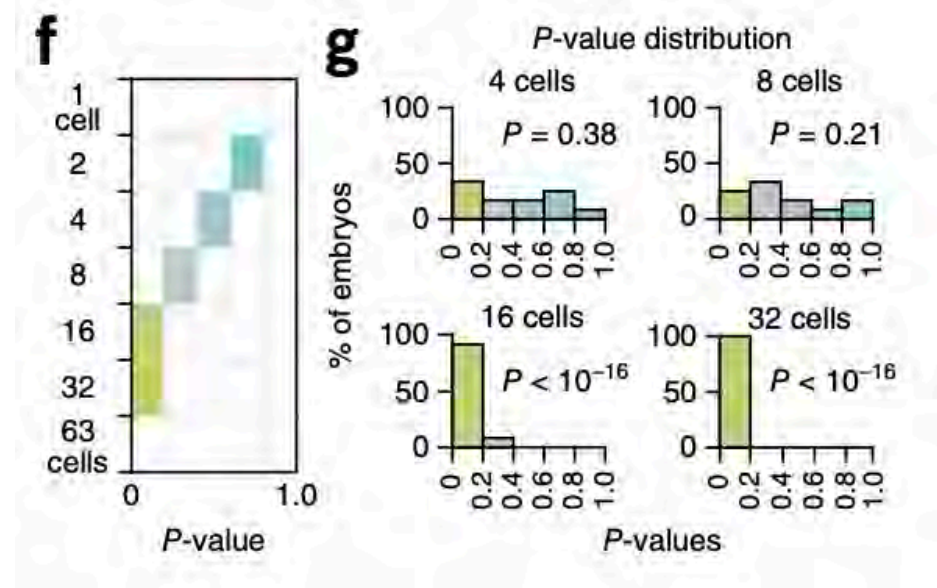
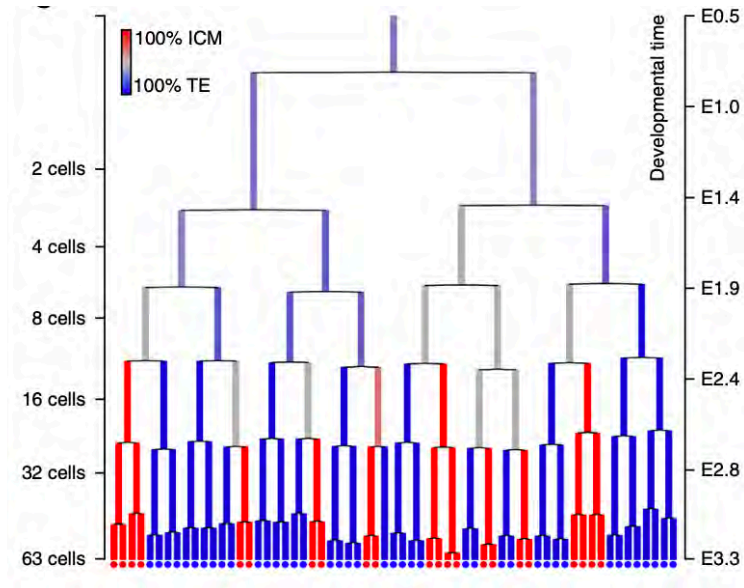


Inverted light-sheet microscope

- Inverted
- Specimen horizontal
- Submerged in water
- Mammalian embryonic development possible
- Many embryos in parallel



Cell fate is biased toward ICM or TE lineage from the 16-cell stage

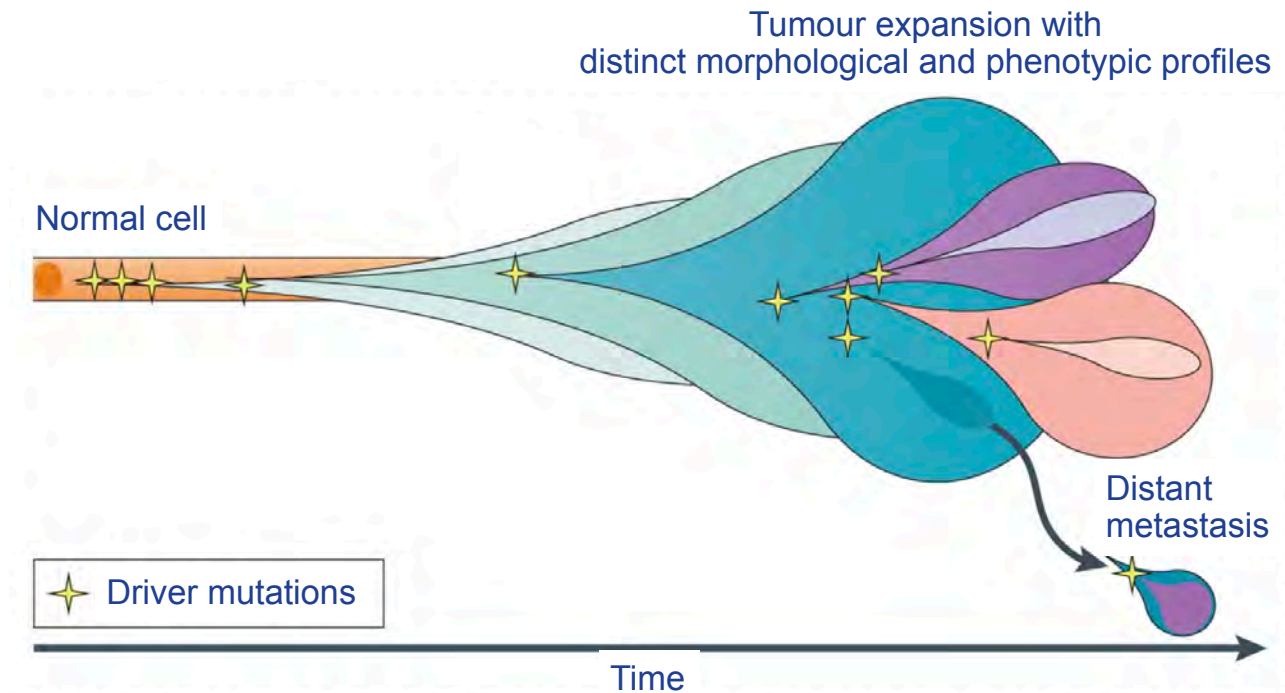


- 16-cell embryo is the earliest stage at which blastomeres are significantly biased in their contributions to TE or ICM cell fate

What we will cover

- Lineage tracing for a systems understanding of disease progression
-

Cancer evolution and tumour heterogeneity in cell carcinoma

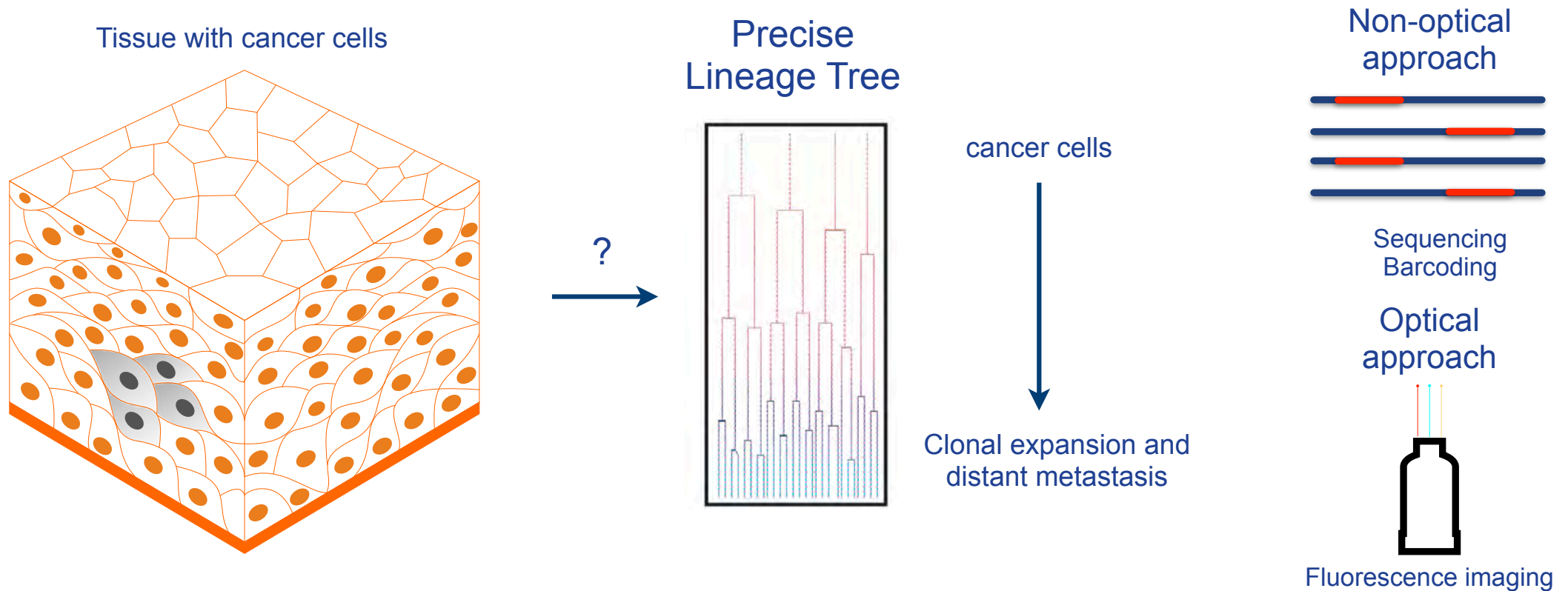


Challenge

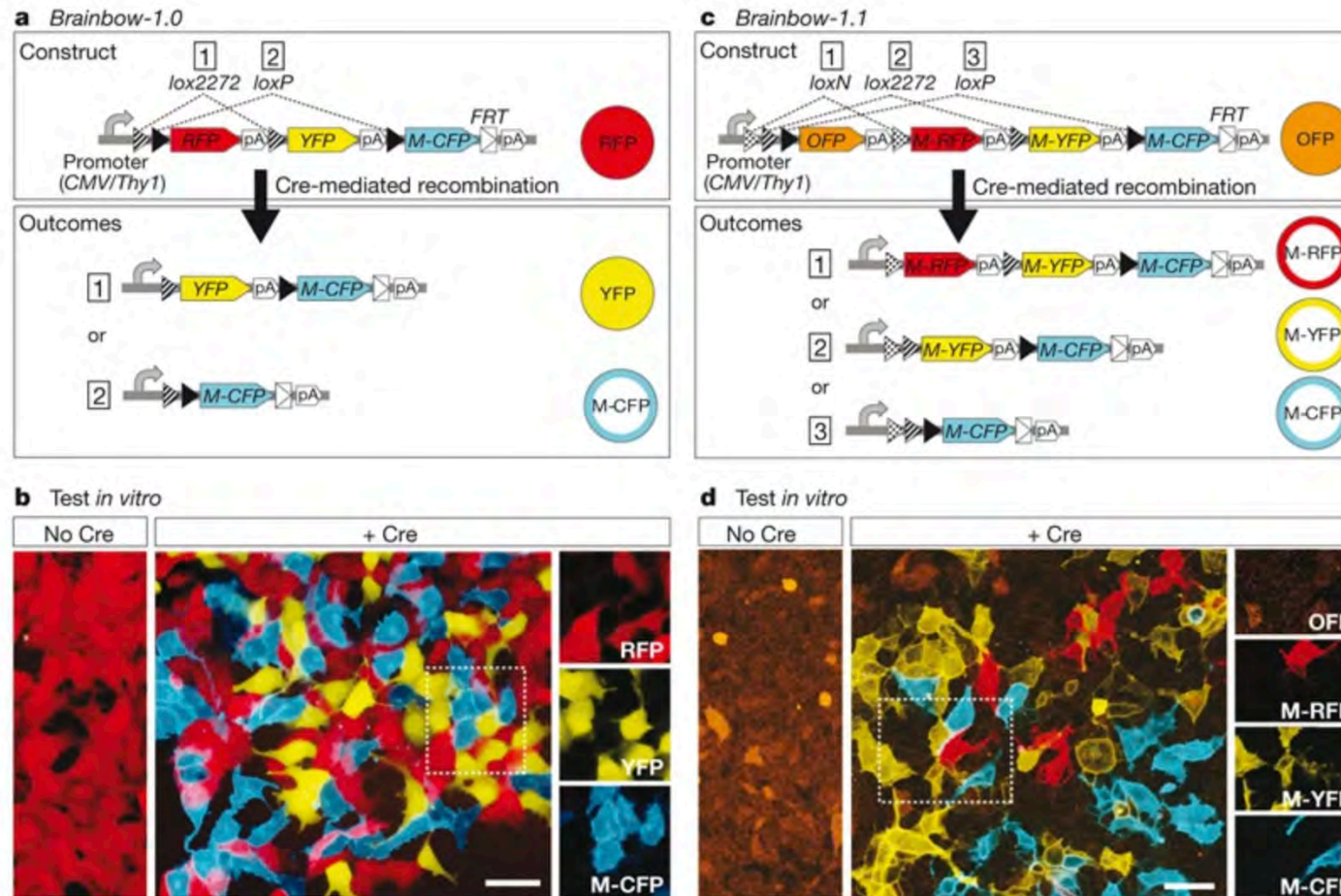
- Tumourigenesis is highly dynamic and involves both genetic and epigenetic changes in the context of a heterogeneous environment
- Cells undergo often complex, non-hierarchical or non-deterministic dynamics
- Cellular histories are inferred rather than measured directly by current single-cell genomics approaches

Spatiotemporal information of malignant cells and cells in the tumour microenvironment allows to display tumour evolution, understand metastasis, monitor and predict the response to therapy

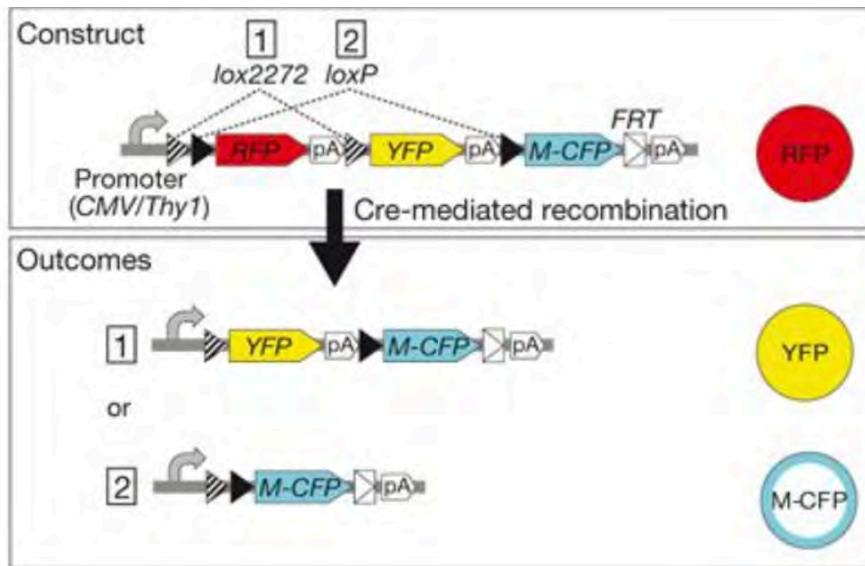
Comprehensive lineage tracing for tumour evolution



Brainbow system: A combinatorial expression of fluorescent proteins



Brainbow system: A combinatorial expression of fluorescent proteins



J. Livet et al. *Nature* **450**, 56 (2007)



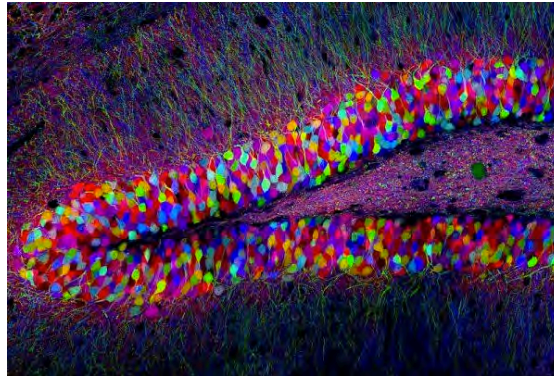
Maximum Color Diversity
= C_{n+k-1}^K

K: Number of insertions
n: Number of fluorescence proteins
C: Combination

		K					
n		1	2	3	4	5	6
	3	3	6	10	15	21	28
	4	4	10	20	35	56	84

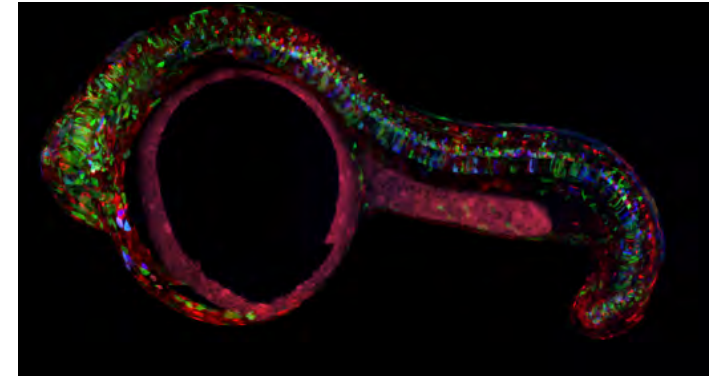
Brainbow systems in various model organisms

2005
Brainbow 1.X and 2.X series
Dentate gyrus - Hilus in the
Brainbow mouse brain



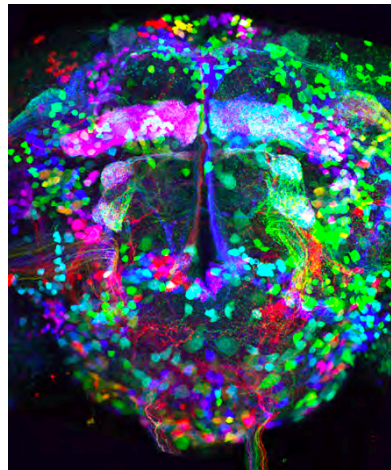
J. Livet, et al. *Nature* **450**, 56 (2007)

2013
Zebrabow

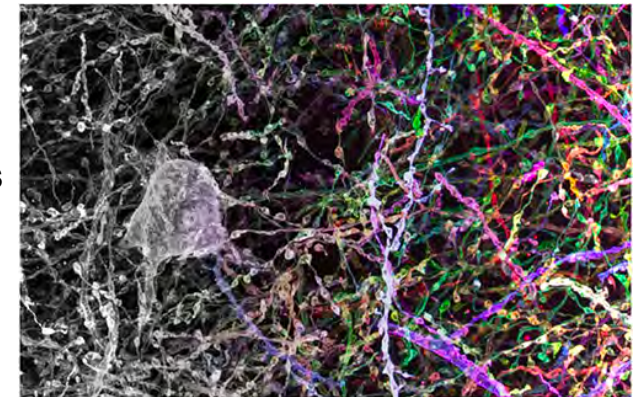


Y.A. Pan, et al. *Development* **140**, 2835 (2013)

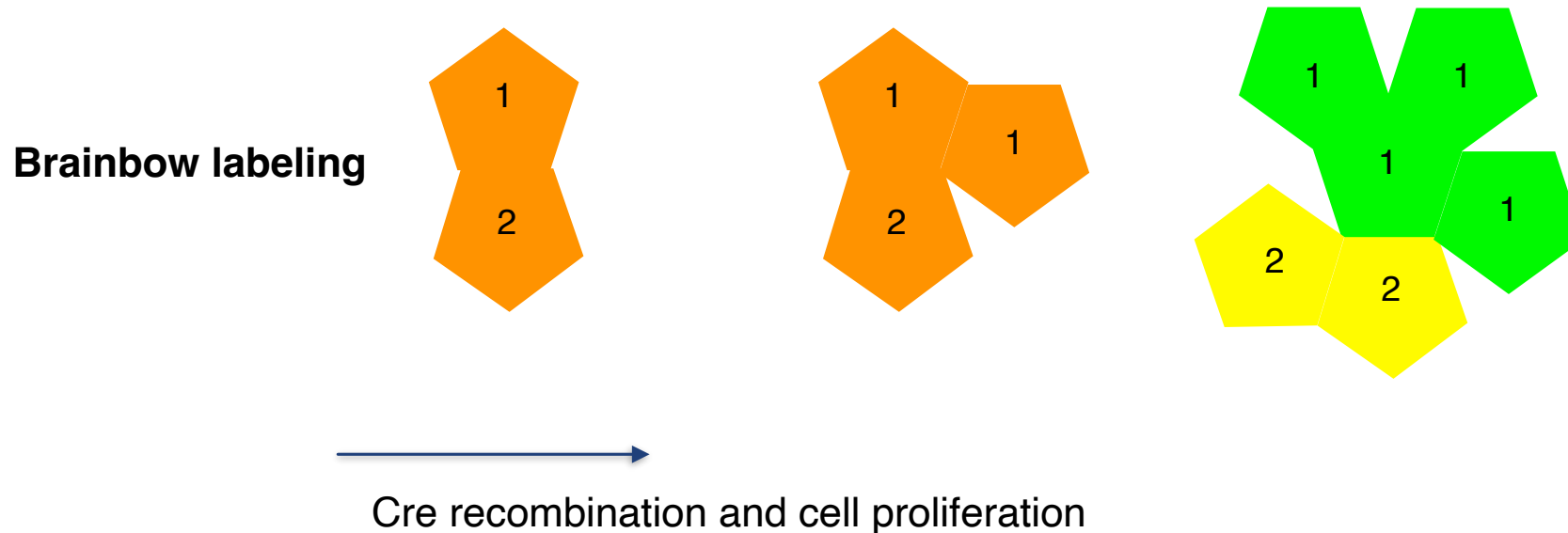
2011
Flybow and dBrainbow



2013
Brainbow 3.X series
Hippocampus

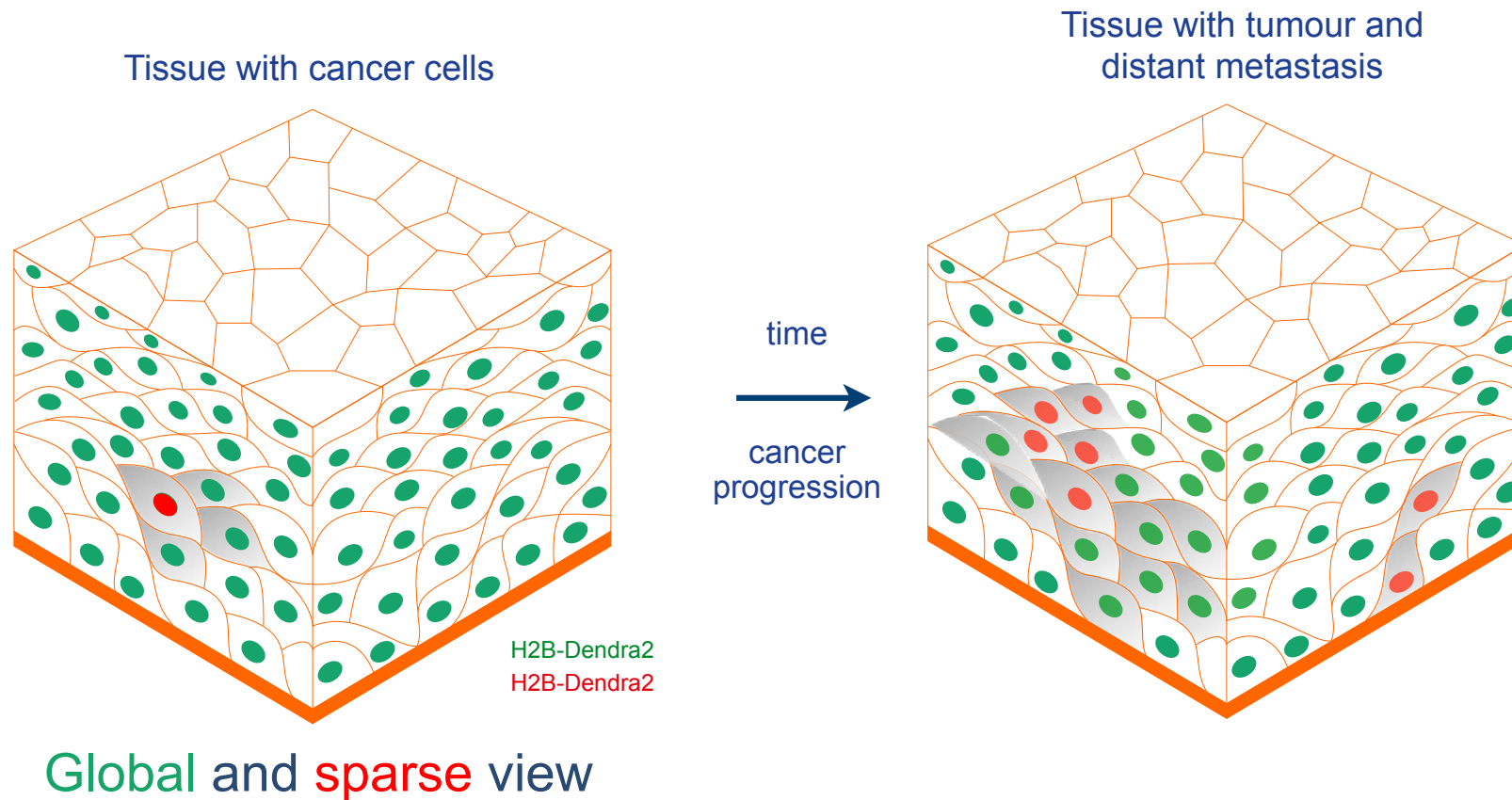


Limitation of the Brainbow system: Slow combinatorial recombination and fluorescent protein maturation



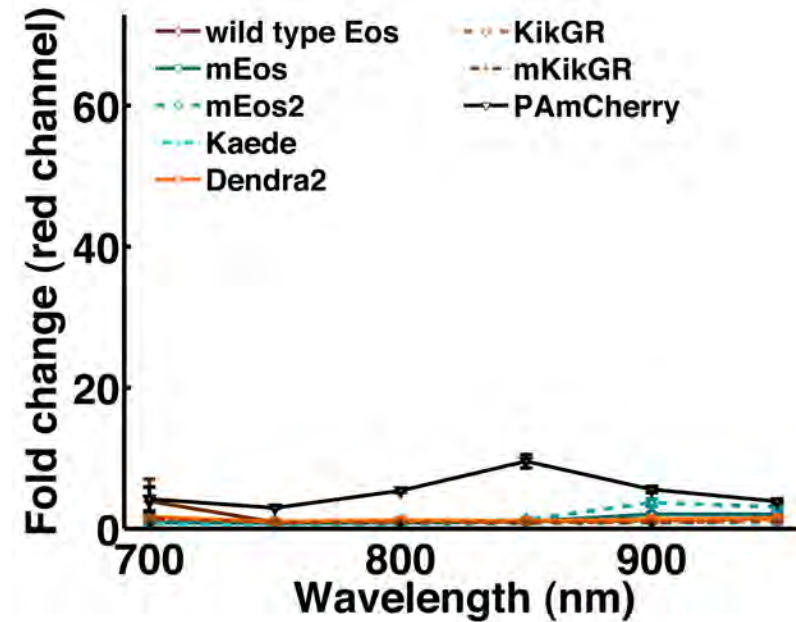
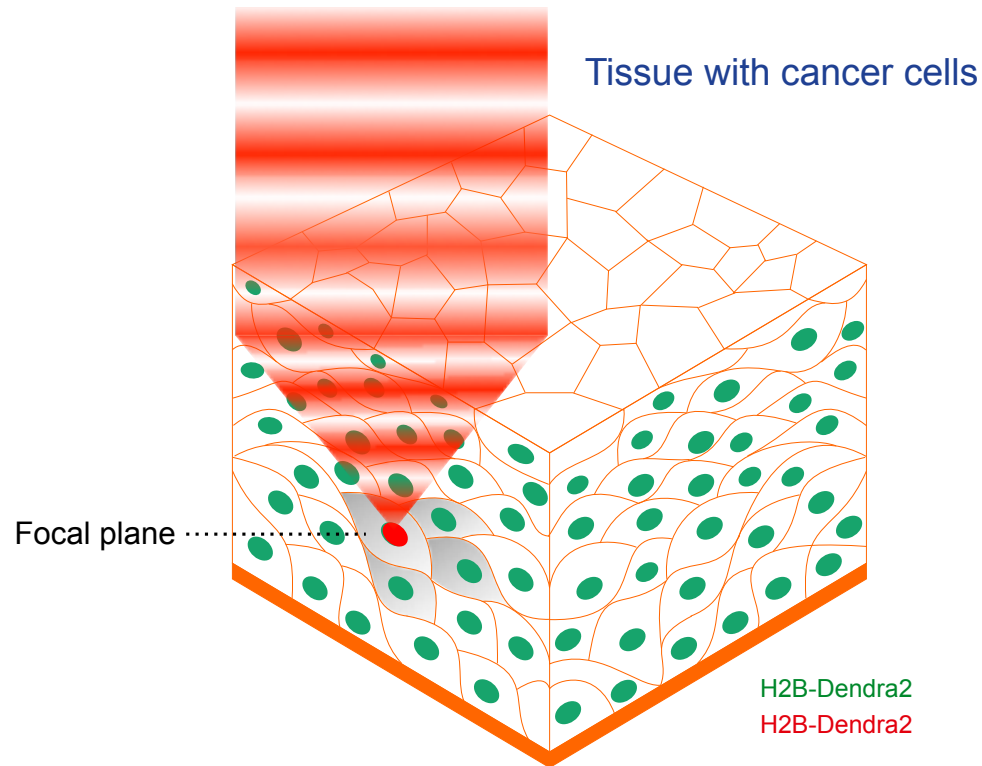
- **Stable steady-state fluorescent labelling is slow**
- **Identification of single-cell origin often not possible**

Comprehensive lineage tracing for tumour evolution using photoconvertible fluorescent proteins



Efficient 2-Photon (2P) photoconversion is not achievable

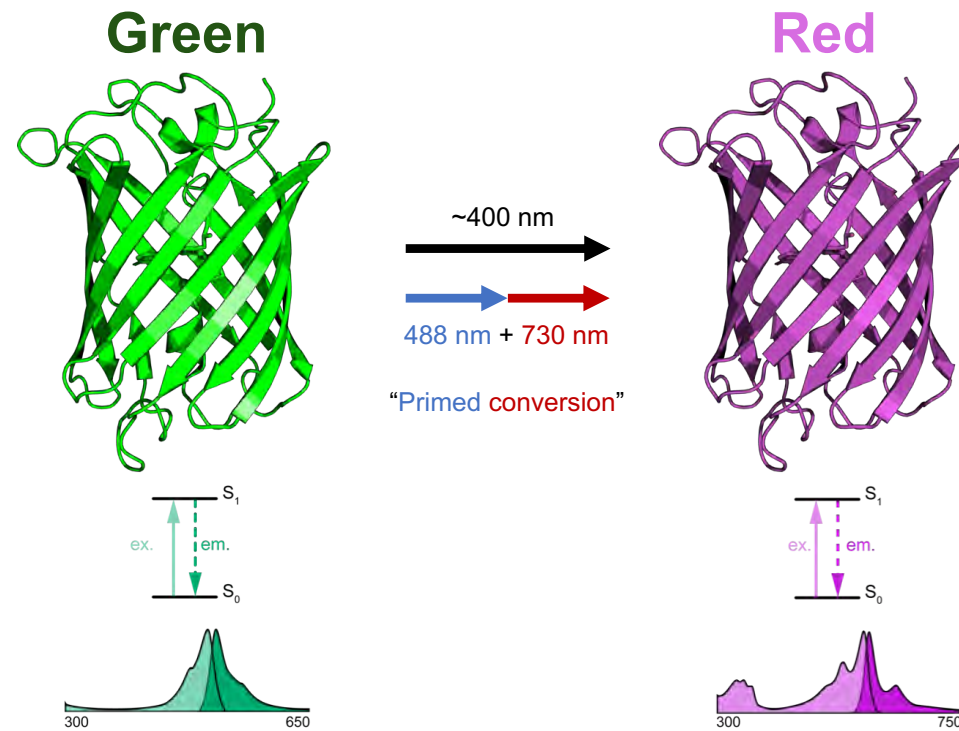
After pulsed near-infrared light illumination



U.S. patent granted (15 May 2018, US 9,971,136).
E.P. patent pending (2014, EP 2014/055669).
W.O. patent pending (2014, WO 2014/147211).

Nature Methods (2015); 12(7):645-48.

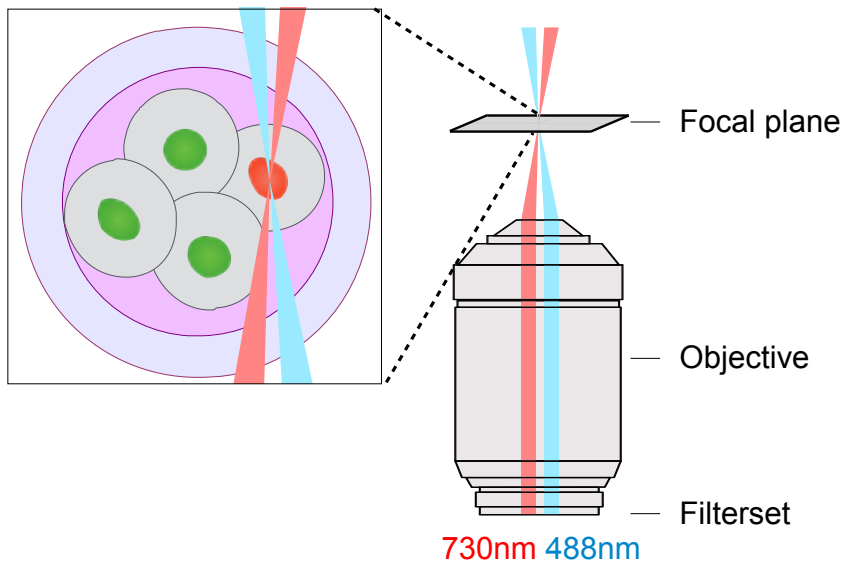
Primed conversion of Dendra2 leads to pronounced photoconversion



U.S. patent granted (15 May 2018, US 9,971,136).
E.P. patent pending (2014, EP 2014/055669).
W.O. patent pending (2014, WO 2014/147211).

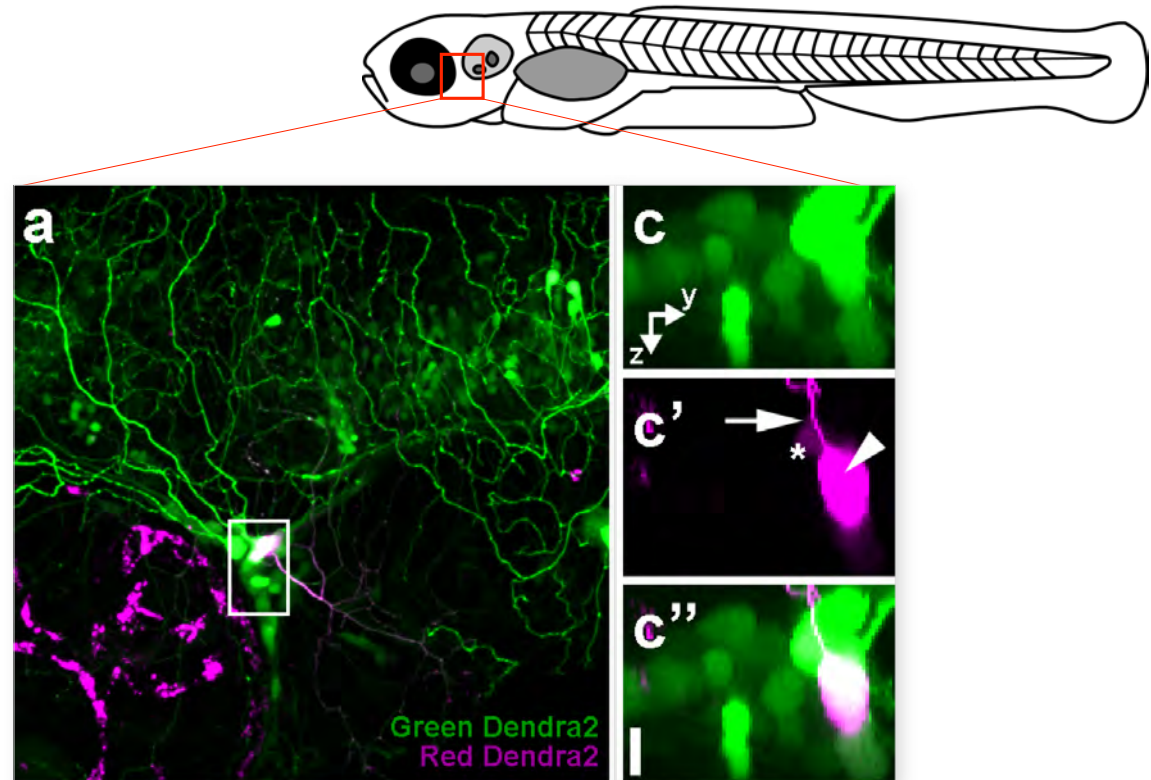
Nature Methods (2015); 12(7):645-48.

Crossed-beam primed conversion at the focus achieves axial confinement in vivo



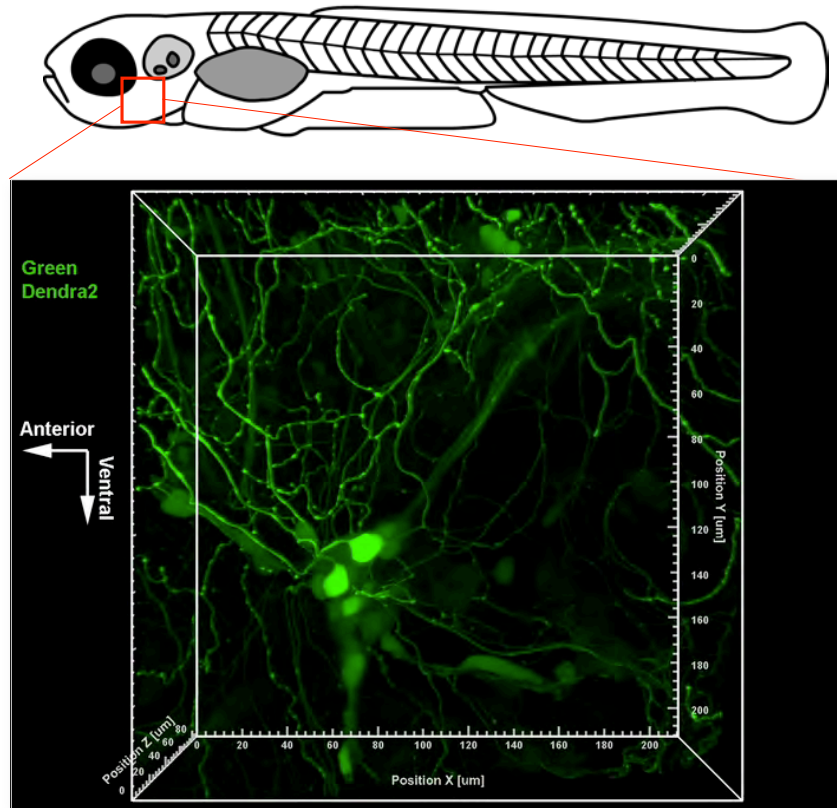
Primed Conversion

Filter available at:



Nat Methods (2015); 12(7): 645-8.
Nat Protoc (2016); 11(12): 2419-2431.
Methods Cell Biol (2016); 133: 125-38.
Chemistry (2018); 24(33): 8268-8274.

Confined primed conversion is an attractive tool in dissecting the structural connectome



U.S. patent granted (15 May 2018, US 9,971,136).
E.P. patent pending (2014, EP 2014/055669).
W.O. patent pending (2014, WO 2014/147211).

Nat Methods (2015); 12(7): 645-8.
Nat Protoc (2016); 11(12): 2419-2431.
Methods Cell Biol (2016); 133: 125-38.
Chemistry (2018); 24(33): 8268-8274.

All photoconvertible proteins can be rendered susceptible to primed conversion

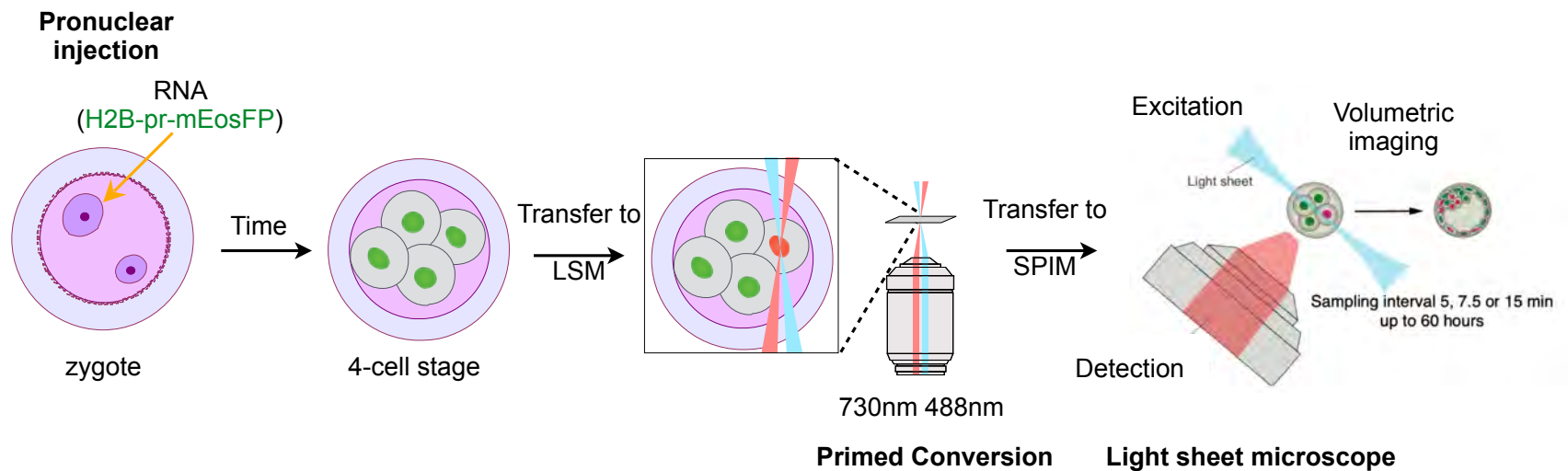
Category	Protein	partial AA sequence	Primed convertible
Monomeric Photoconvertible fluorescent proteins (PCFPs)	Dendra2	<u>HYGN</u> RVFTKYPED	Yes
	mClavGR	<u>HYGN</u> RVFTKYPED	Yes
	mMaple	<u>HYGN</u> RVFTKYPED	Yes
	mEosFP	HYGNRVFVEYDPH	No
	pr-mEosFP	HYGNRVFTKYPED	Yes
	mEos2	HYGNRVFAKYPDN	No
	pr-mEos2	HYGNRVFTKYPDN	Yes
	mEos3.1	HYGNRVFAKYPDN	No
	pr-mEos3.1	HYGNRVFTKYPDN	Yes
	mEos3.2	HYGNRVFAKYPDN	No
	pr-mEos3.2	HYGNRVFTKYPDN	Yes
	mEos4a	HYGNRVFKYPDN	No
	pr-mEos4a	HYGNRVFTKYPDN	Yes
	mEos4b	HYGNRVFKYPDN	No
	pr-mEos4b	HYGNRVFTKYPDN	Yes
	mKikGR	HYGNRVFEYPEE	No
	pr-mKikGR	HYGNRVFTYPEE	Yes
multimeric PCFPs	Kaede	HYGNRVFAKYPDH	No
	pr-Kaede	HYGNRVFTKYPDH	Yes
transcriptional activators	PhoCI	HYGNRVFTKYPRG	Yes
Calcium integrators	CaMPARI	HYGNRVFKYPDN	No
	pr-CaMPARI	HYGNRVFTKYPDN	Yes
Calcium sensors	GR-GECO1.1	HYGNRVFTKYPED	Yes
	GR-GECO1.2	HYGNRVFTKYPED	Yes



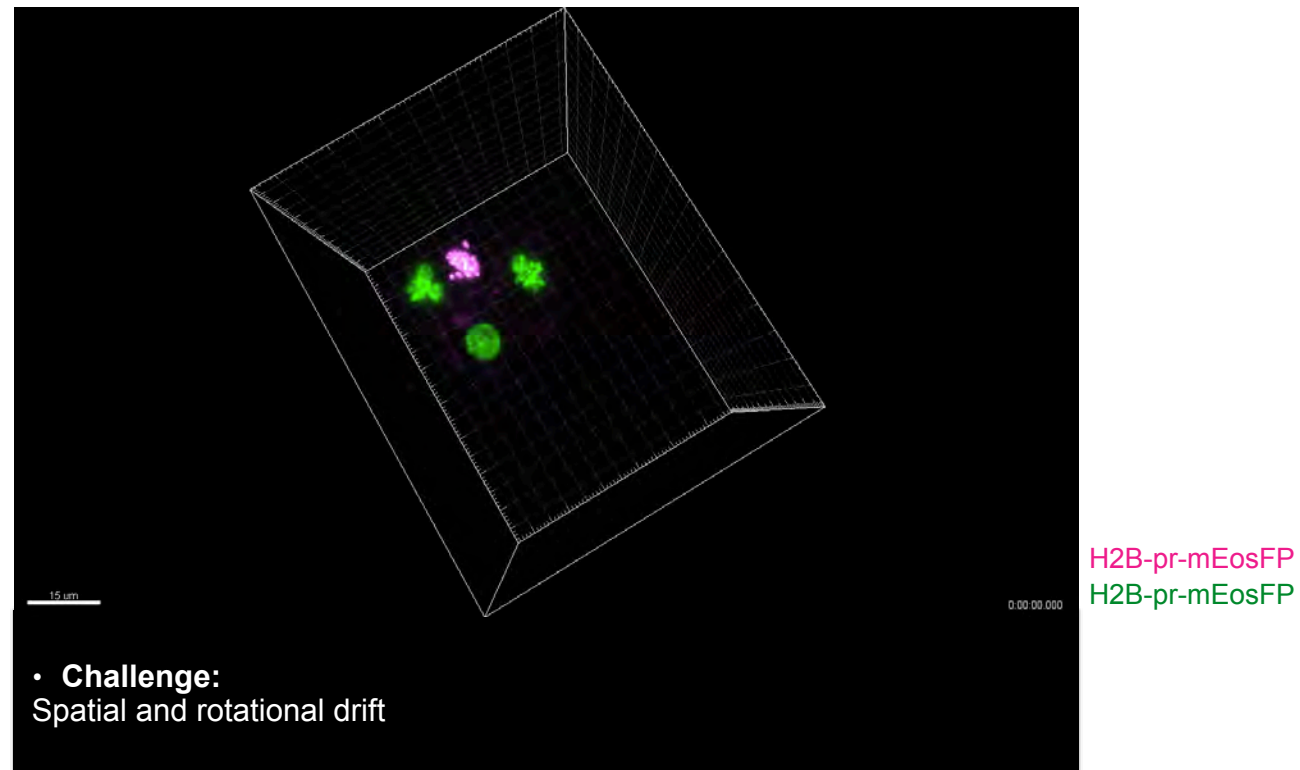
Plasmids available at:



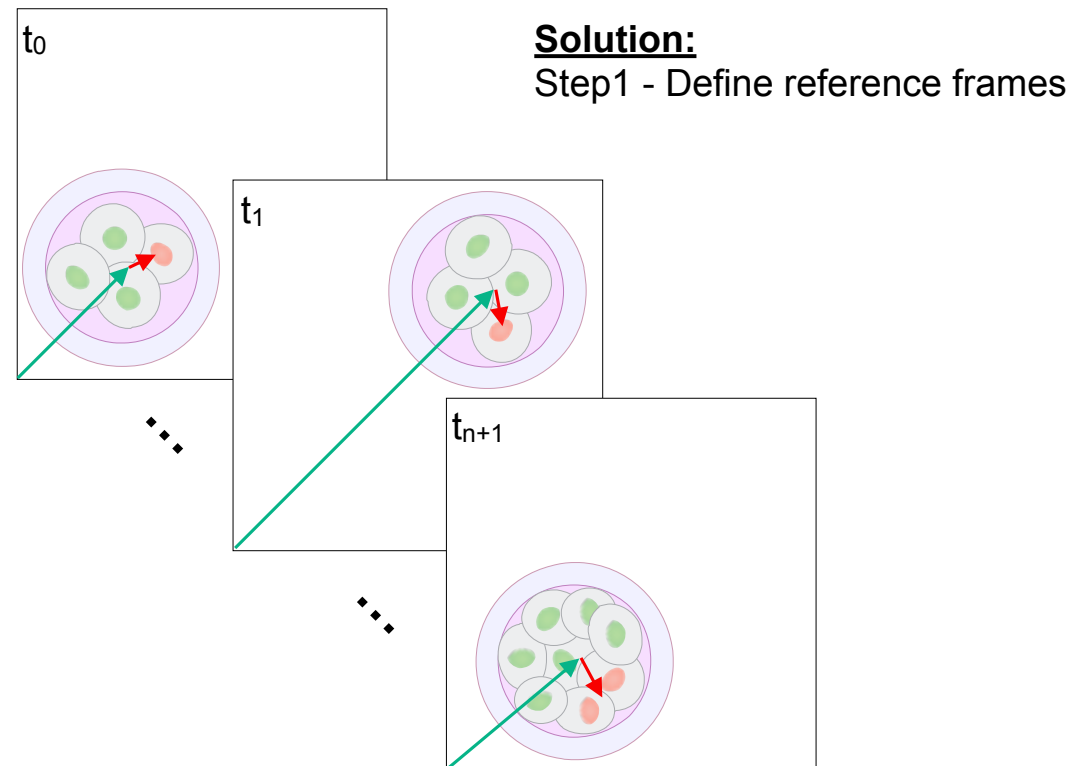
Tracking cells in the early mammalian embryo using primed conversion and volumetric imaging



Tracking cells in the early mammalian embryo using primed conversion and volumetric imaging



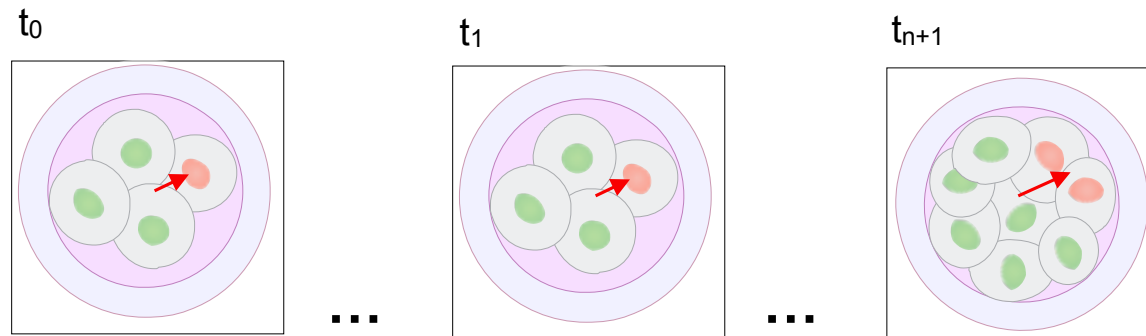
primed Track: High fidelity cell tracking in mouse embryos using primed conversion



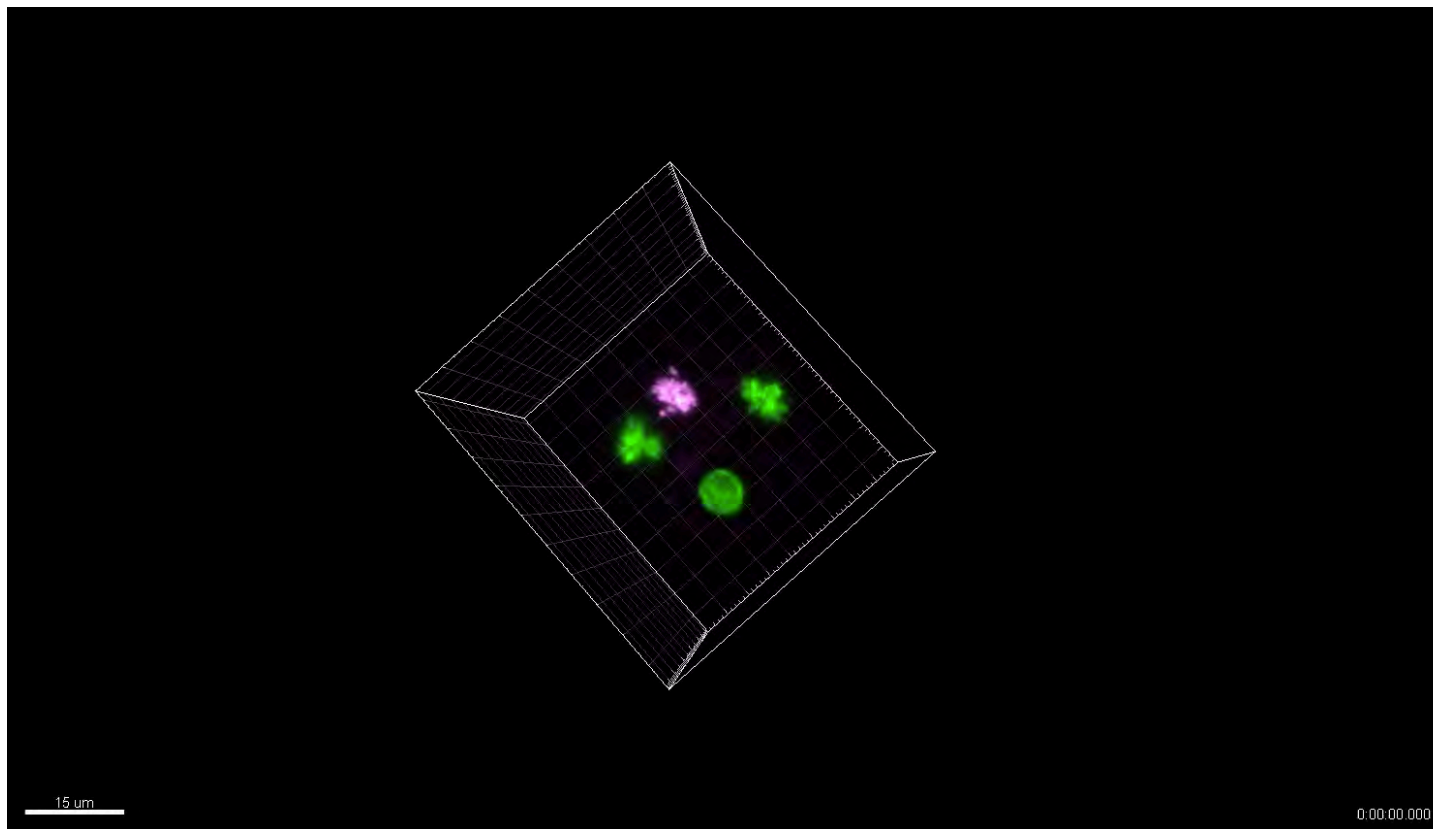
primed Track: High fidelity cell tracking in mouse embryos using primed conversion

Solution:

Step 2 - Rotate, sample, and crop



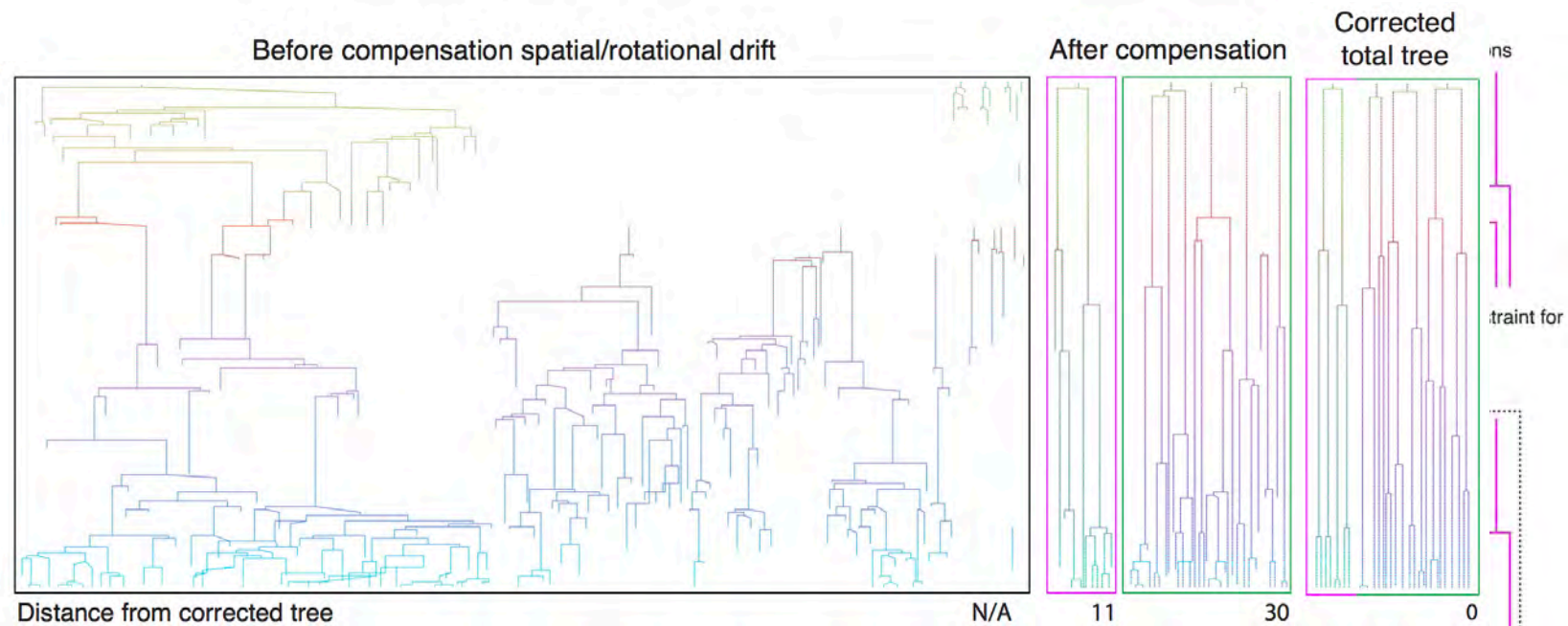
primed Track: High fidelity cell tracking in mouse embryos using primed conversion



H2B-pr-mEosFP
H2B-pr-mEosFP

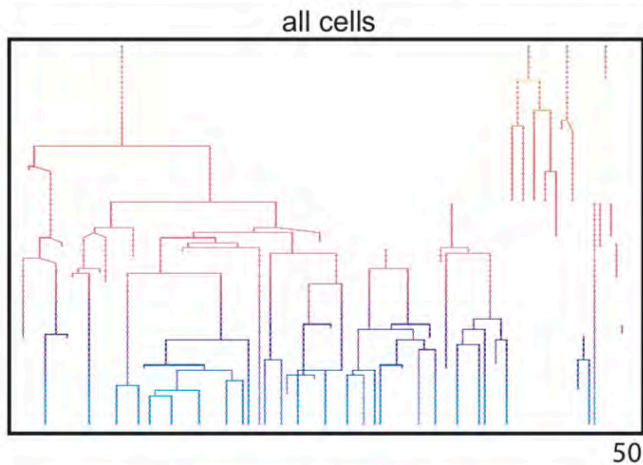
eLIFE (2019); 21(8):e44491.

primed Track: High fidelity cell tracking in mouse embryos using primed conversion

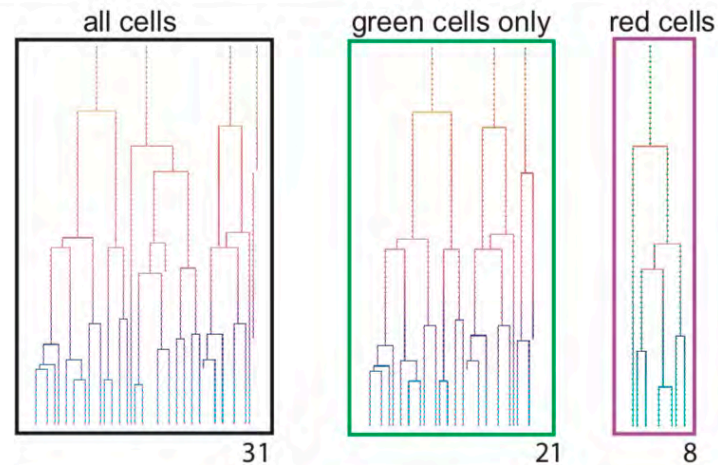


primed Track: High fidelity cell tracking for embryos imaged every 40 minutes

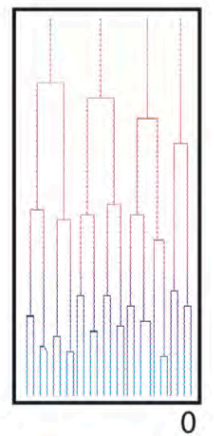
Before compensation spatial/rotational drift



After compensation spatial/rotational drift



Corrected
total tree



Algorithm available at:

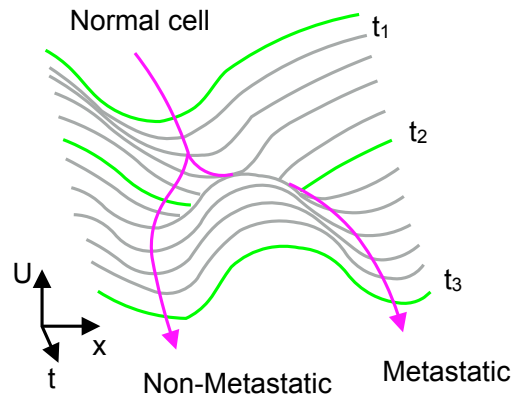


Comparison of Primed Conversion and Brainbow system

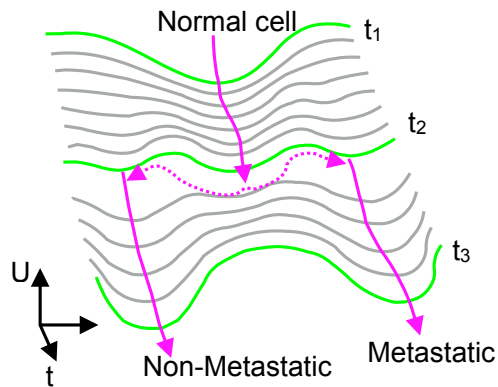
	Brainbow	Primed Conversion
Nature of cell labelling		
Analysis of single cell dynamic		

Dynamic view of achieving heterogeneity in tumour evolution

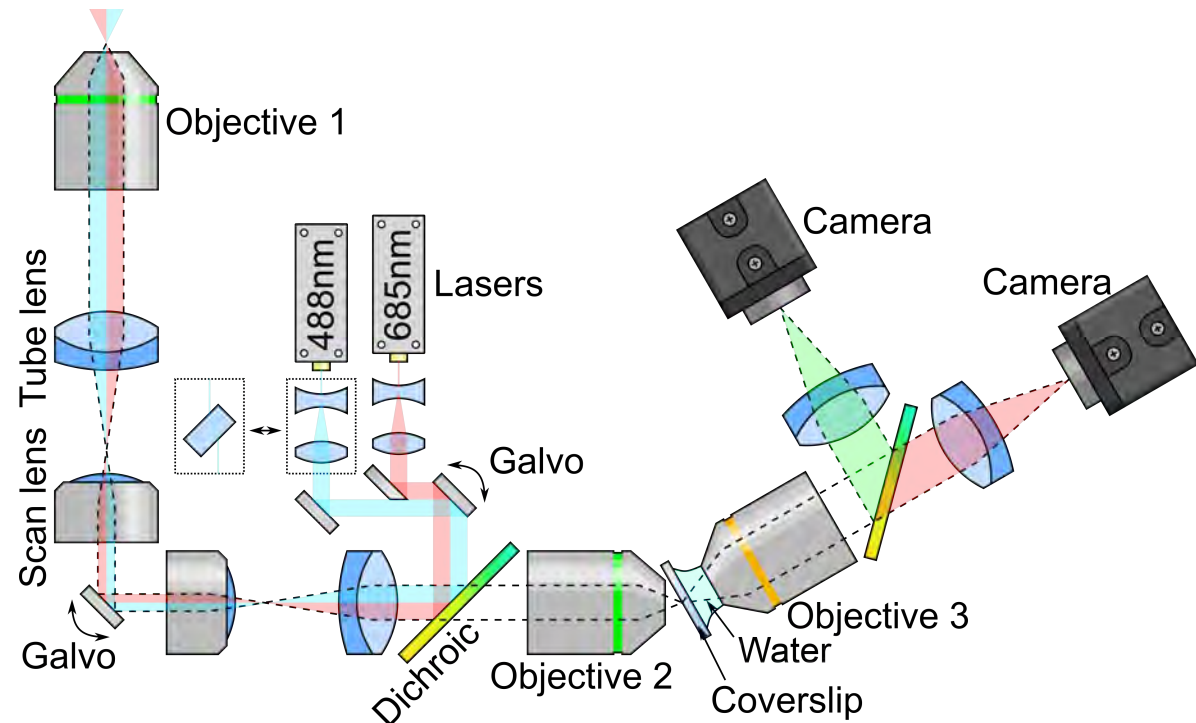
Supercritical pitchfork bifurcation



Subcritical pitchfork bifurcation



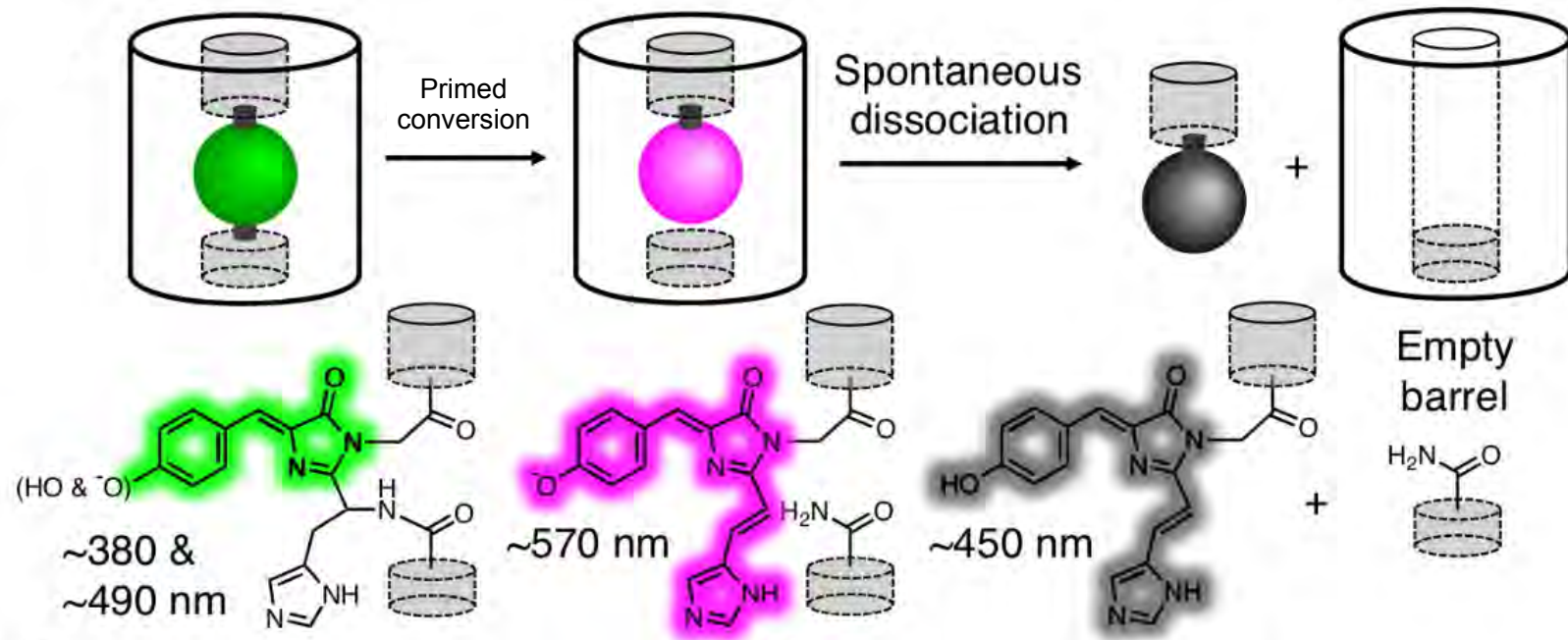
Inverted primed conversion light-sheet microscope



track emergence of heterogeneities in tissue or cancer organoids with subcellular resolution and unlimited numbers of cell divisions

In preparation.

Optogenetic perturbation-control with a photoconvertible PhotoCleavable protein (PhoCI)



Optogenetic manipulation of Cre localisation and gene expression with PhoCI and primed conversion

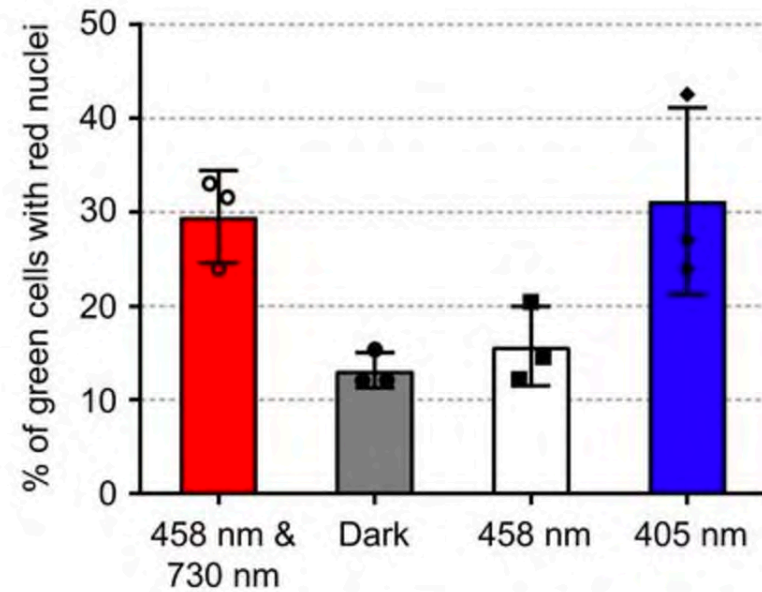
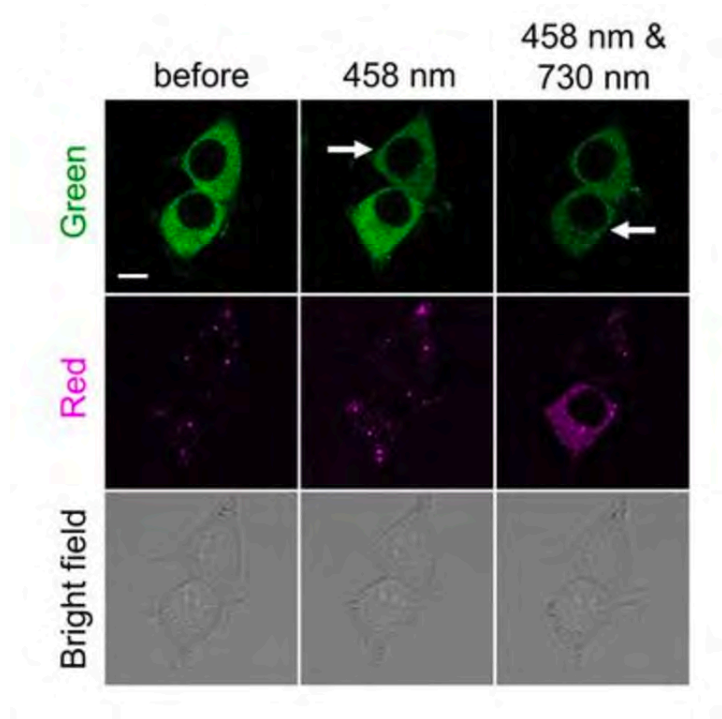
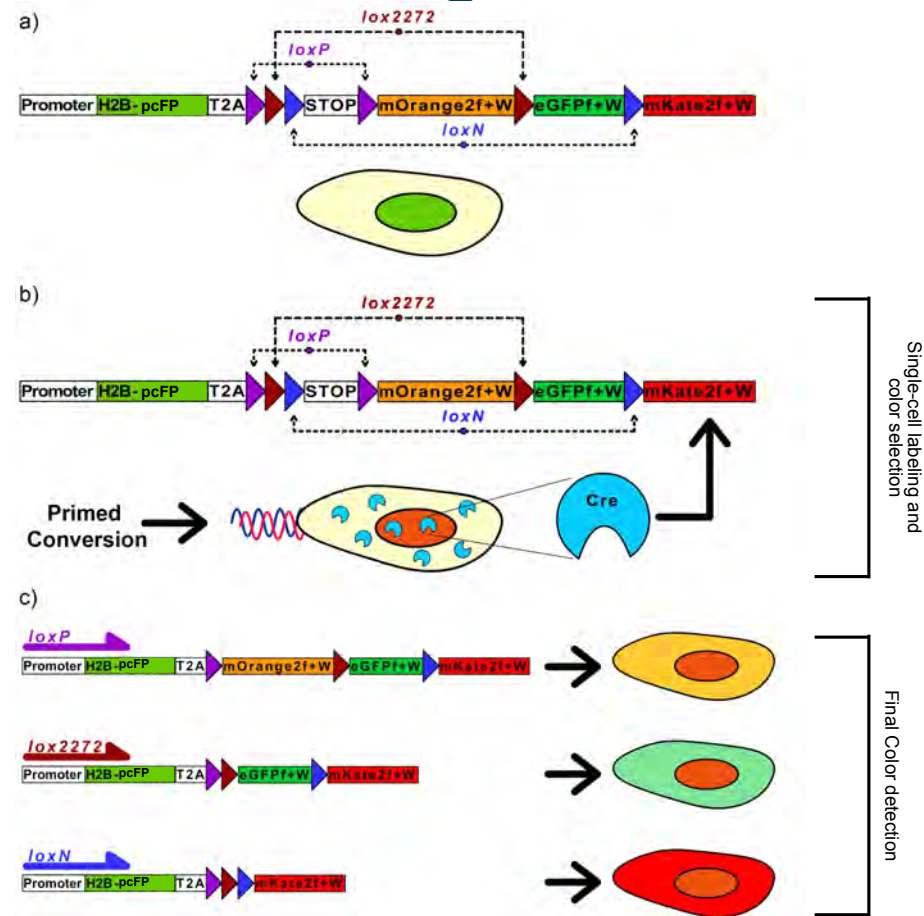
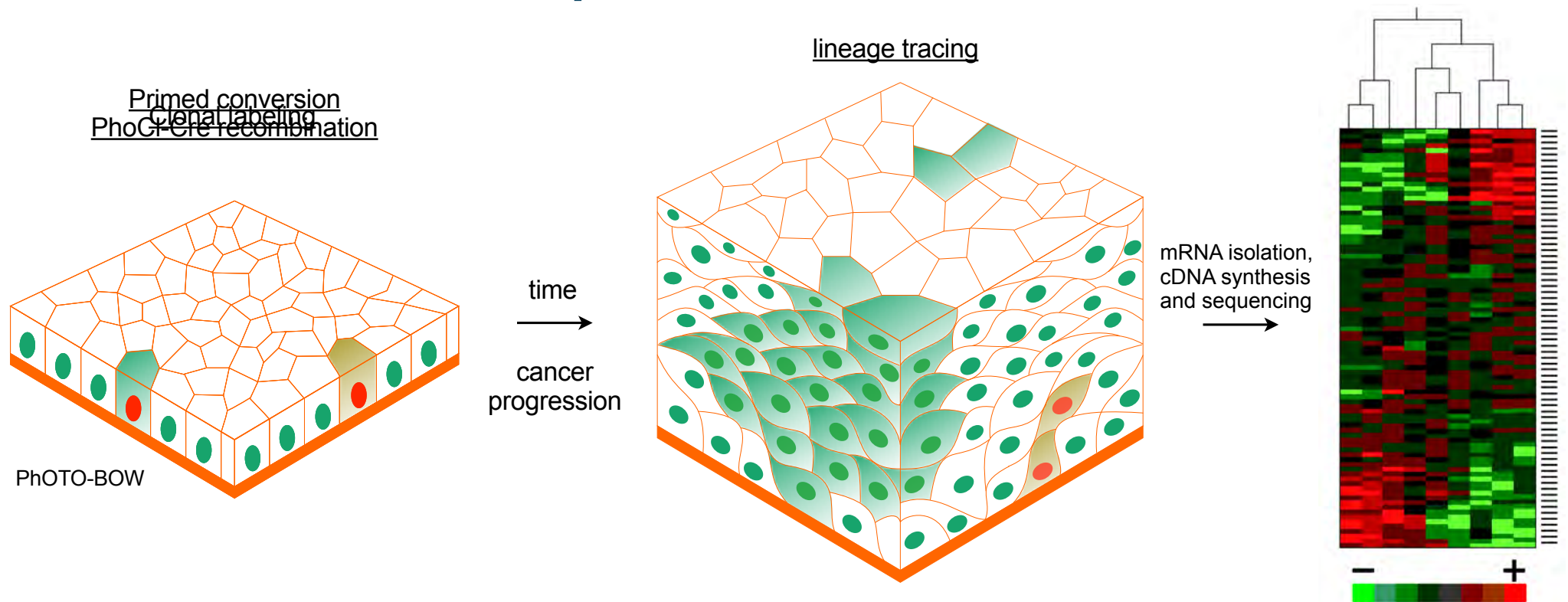


PhOTO-BOW: combining single-cell tracking and clonal lineage-tracing



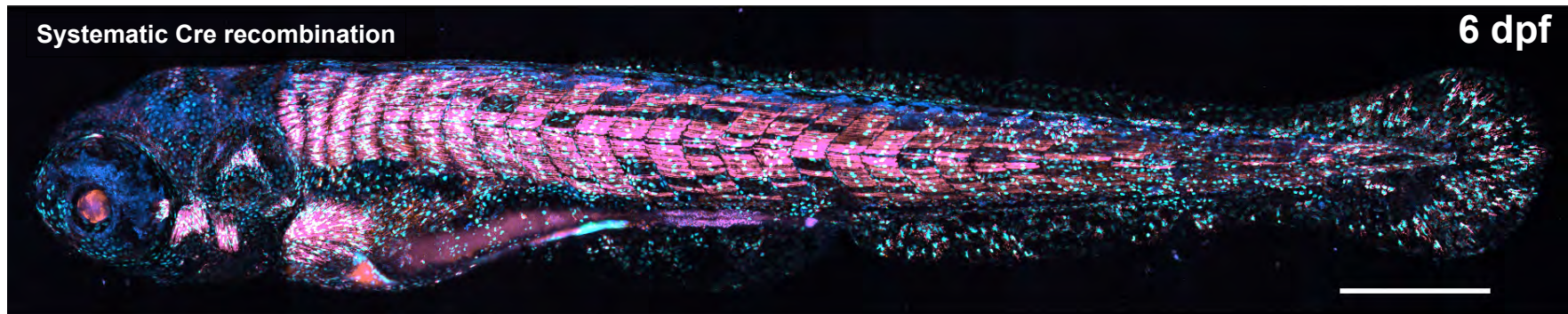
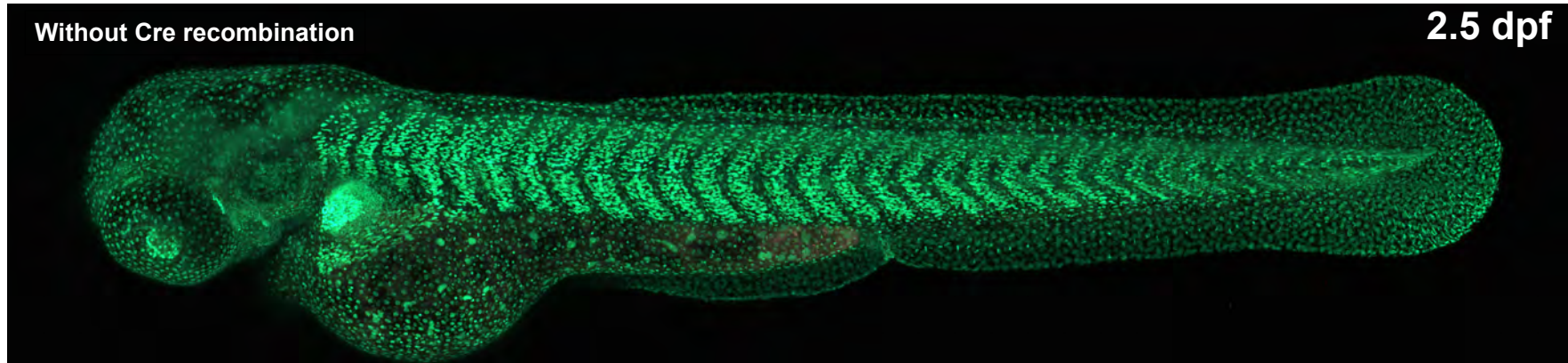
In preparation.

PhOTO-BOW: reconstructing lineage trees throughout an organism and across developmental and disease time



In preparation.

PhOTO-BOW: combining single-cell tracking and clonal lineage-tracing



In preparation.

Thank you!